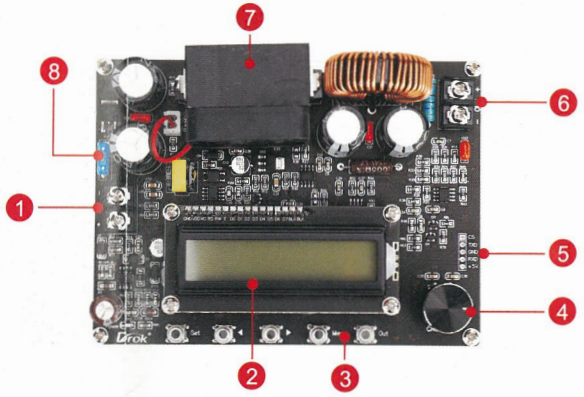
200683/200684

DROK DC CNC Buck Module

Parameters

Model	200863	200864
Screen	LDC 1602	
Input Voltage	10V~100V	
Output Voltage	0 ~ 62V	0 ~ 90V
Output Current	0 ~ 12A	0 ~ 10A
Input Protection	Reverse Connection Protection	
Output Protection	Over Voltage Protection (OVP) Over Current Protection (OCP) Over Power Protection (OPP)	
Output Power	0 ~ 720W	0 ~ 900W
Voltage Regulation	10mV	
Current Regulation	10mA	
Output Ripple	< 100mVpp (Input 48V, Output 24V, Current 5A)	
100Hz Fluctuation	< 1/10000	
Voltage and Current Display	10mV, 10mA	
Voltage Display Error	±1%+20mV	
Current Display Error	±2%+20mA	
Response Time	< 50ms	
Heat Dissipation	Onboard Heat Sink and Cooling Fan	
Fan Start-up Condition	Output Current>1.5A	
Operating Ambient	0~60℃	
Storage Ambient	-20~70℃	
Operating Environment	Indoor Use, Maximum Humidity: 80%	
Size (Length×Width×Height)	135×95×38(mm)/5.3x3.7x1.5(inch)	

Parts of the Module



NO.	Part	NO.	Part
1	Input Port	5	Communication Port
2	LCD Display	6	Output Port
3	Operation Buttons	7	Heat Sink and Cooling Fan
4	Rotary Encoder	8	15A Automotive Fuse

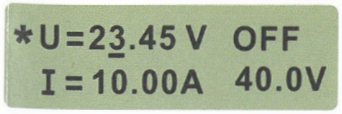
Operation Instruction

- **Wiring:**

The input (IN) and output (OUT) are connected correctly to ensure that the input voltage is within the required range. Reverse connection is forbidden!

- **Voltage and Current Values Setting:**

(1) Setting Voltage Value



Press the "Set" to switch "*" to the above position, then press ◂button or ▸button to move the cursor and rotate the encoder to adjust the voltage value.

(2) Setting Current Value

```
U=23.45 V  OFF
*I=10.00A  40.0V
```

Press the "Set" to switch "*" to the above position, then press ◂button or ▸ button to move the cursor and rotate the encoder to adjust the current value.

(3) After setting the parameter value, press the "Out" button to output. If the parameter value is changed, the system will save the parameter value. Next time when restart, default value will be the last saved value.

- **Start Output:**

After setting output voltage and current value, press "Out" to turn on or turn off output.

```
CV 23.45 V   8.00A
IN = 40.0V 187.6W
```

- **Adjust Voltage and Current Value in the Output State:**

In the output state, press "Set" to switch setting voltage or current, press ◂button or ▸ to move the cursor and rotate the encoder to adjust the voltage value.

- **Display Capacity and Time Function:**

```
CV 23.45 V   8.00A
IN = 40.0V 187.6W
```

```
CV 23.45 V   8.00A
  40.00AH 05:00:00
```

In the output state, press "Mer" to switch display function.

- **System Parameter Setting:**

This Module has 4 sets of parameters:

Input Low Voltage Protection (LVP):
After turning on LVP function, when input voltage is lower than the set LVP voltage, module will automatically turn off the output.

Over Voltage Protection (OVP):
After turning on OVP function, when output voltage is higher than the set OVP voltage, module will automatically turn off the output.

Over Current Protection (OCP):
After turning on OCP function, when output current is higher than the set OCP current , module will automatically turn off the output.

Self-tuning Regulator (STR):
After turning on STR, in the LVP output off state, when input voltage is higher than the set STR voltage, module will auto matically turn on the output.

(1) Setting LVP / OVP / OCP / STR

- LVP / OVP / OCP / STR setting method are similar. Take setting LVP as an example, in the output off state, press the "Set", when "*" disappears (as shown below) , clockwise rotate the encoder to enter the parameter setting interface.

```
U=23.45 V  OFF
I=10.00A  40.0V
```

```
System Set
LVP  10.0V  OFF
```

- Press "Out" to turn on or turn off protection functions, press "Set" to display cursor, press ◂button or ▸ button to move the cursor and rotate the encoder to adjust the parameter value.

```
System Set
LVP  10.0V  ON
```

- Press "Set" to make the cursor disappear , then rotate the encoder clockwise to switch the kinds of parameter, rotate the encoder anti-clockwise to exit the parameter setting.
- Then press "Mer" to save setting.

(2) Setting ATO Function

Automatic Output (ATO):
After turning on ATO function, the module will automatically turn on output when power on, there's no need to press "Out" to start output.
Enter the Automatic Output (ATO) function, press "Out" to turn on or turn off the ATO function, then press "Mer" to save setting.

```
System Set
ATO        OFF
```

```
System Set
ATO        ON
```

Notes

This module has Input Low Voltage Protection (LVP) and Self-tuning Regulator (STR), if both value are set and the functions are turned on, the module will stop output when the input voltage is lower than LVP. Moreover, the module will restart output again while the input voltage is higher than STR.

- **Communication Protocol:**

Baud rate: 9600Bps; Address bits 0-26 correspond to a-z

- **Write data instruction:**

Instruction format: a(address bits)+w+(command character) +XXXX(data string)+s(end character)

- Set output voltage instruction:
  awu1134s Set the output voltage to 11.34V
  awu0134s Set the output voltage to 13.4V
- Set output current instruction:
  awi0100s Set the output current to 1.00A
  awi1050s Set the output current to 10.50A
- Set on/off state:
  awo0001s Turn on the output
  awo0000s Turn off the output
- Call saved parameter instruction:
  awm0s Call the parameter saved in address 0
  awm5s Call the parameter saved in address 5
- Save parameters:
  aws0s Save the set voltage and current values to address 0
  aws5s Save the set voltage and current values to address 5

- **Read data instruction:**

Instruction format: a(address bits)+r+(command character) +s(end character)

- Read output voltage instruction:
  arus Returns data: #ru0004000s Actual output voltage 40.00V
- Read output current instruction:
  aris Returns data: #ri0000500s Actual output current 5.00A
- Read residual capacity instruction:
  arcs Returns data: #rc0000050s Actual capacity 0.50AH
- Read set voltage instruction:
  arvs Returns data: #rv0004000s Actual set voltage 40.00V
- Read output status instruction:
  aros Returns data: #rv0000001s Output on; #rv0000000s Output off